Transit Data Conference 2026 - Technical Report

Improving Data Quality in TTC's Open Data: TTC Bus Delay Data & TTC Streetcar Delay Data

The Delayticans

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1 Problem Statement

Before we actually come up with this topic, we tried to build a model to predict the delay of the TTC buses and streetcars based on the current weather data. However, we found that the data is not enough to build a good model. Hence, we decided to focus on analyzing and improving the data quality instead.

When we tried to clean the data, we found that the data is not consistent. First of all, we can ignore some minor problems such as inconsistent date and time format. Some rows have missing values, a few rows have an incorrect value, and some rows have questionable values such as negative values and zeros. Additionally, there is a noticeable ratio between the minutes of delay and the minutes of gap in some delay events, usually 1 : 2. Finally, in some delay events, the minutes of delay is equal to the minutes of gap, which raises questions about the accuracy of the data.

The images below are examples that show the data quality issues:

Date	Line	Time	Day	Location	Incident	Min Delay	Min Gap	Bound	Vehicle
31-Dec-23	510	22:00	Sunday	SPADINA AND COLLEGE	General Delay	0	0	N	4477
31-Dec-23	505	22:02	Sunday	DUNDAS AND YONGE	General Delay	0	0	W	0
31-Dec-23	509	22:03	Sunday	UNION STATION	General Delay	0	0		0
31-Dec-23	505	22:19	Sunday	LESLIE CARHOUSE	Operations	10	20		4537

Figure 1: Zeros and nulls appeared in TTC's 2013 streetcar delay data.

Report Date	Route	Time	Day	Location	Incident	Min Delay	Min Gap	Direction	Vehicle
10-Oct-18	29	3:00:00 PM	Wednesday	Dundas and Bloor.	General Delay	20	26	N/B	
10-Oct-18	37	3:00:00 PM	Wednesday	Islington stn	General Delay	20	30	B/W	7967
10-Oct-18	929	3:00:00 PM	Wednesday	Dundas and Dufferin	General Delay	20	29	N/B	
10-Oct-18	952	3:00:00 PM	Wednesday	Lawrence Station to Terminal 3	General Delay	-54	66	B/W	

Figure 2: Negative values appeared in TTC's 2018 bus delay data.

Report Date	Route	Time	Day	Location	Incident	Min Delay	Min Gap	Direction	Vehicle
01-Jan-18	28	1:45:00 AM	Monday	Bayview and Pottery	Emergency Services	15	30	76647	8202
01-Jan-18	39	1:45:00 AM	Monday	NECR	Mechanical	10	20	W/B	1768
01-Jan-18	120	1:58:00 AM	Monday	Wilson Stn	Mechanical	30	60	S/B	8529
01-Jan-18	28	2:00:00 AM	Monday	Bayview and Pottery Rd	Diversion	0	0	S/B	8306

Figure 3: Incorrect values and the noticeable ratio between Min Delay and Min Gap appeared in TTC's 2018 bus delay data.

Therefore, this report will visualize the data quality issues and provide some suggestions for improving the data quality. As the result, future researchers can access to well-structured data and make better decisions.

2 Methodology

This study is quantitative in nature. First of all, we downloaded the raw data from Toronto's Open Data and merged them into a single CSV file in each year. We collected all the delayed events in buses and streetcars from 2014 to April 2026. Then, we used Python to clean the data and visualize the data quality issues. Specifically, we used the following libraries:

- `pathlib`: To merge all the raw data into a single CSV file in each year and manage the file paths.
- `matplotlib, numpy`: To visualize the data quality issues.

After that, we analyzed the rate of rows with zero or null values in different columns such as Min Delay, Min Gap, Bound, and Vehicle, across the years. In addition, we calculated the rate of rows with the value of Min Delay and Min Gap in a generic ratio such as 1 : 2 or 1 : 1. At the moment, through all the years, there are 776,446 delayed events in buses and 163,010 delayed events in streetcars in total. We used a combination of pie charts bar graphs, and line graphs to visualize the data quality issues.

Pie Charts

They show the proportion of rows with zero or null values in multiple columns for each year, allowing us to identify potential data quality issues in specific years. Each pie chart has a title that shows the percentage of rows that has at least one zero, negative, or null value.

Bar Graphs and Line Graphs

They show the yearly trend of the proportion of rows with zero or null values across the 2014–2025 period, as well as the ratio of Min Delay and Min Gap, allowing us to identify any patterns or anomalies in the data quality over time and across different vehicle types.

3 Data Sources

The data for this analysis was sourced from Toronto's Open Data portal, specifically the TTC Bus Delay Data and TTC Streetcar Delay Data. These datasets cover the years 2014 through April 2026, providing a comprehensive view of delay events over this period. Each dataset includes multiple columns, which are Date/Reported Date, Line/Route, Time, Day, Location, Incident, Delay code *(since 2025)* Min Delay, Min Gap, Bound/Direction, and Vehicle. The data is collected and updated by the Toronto Transit Commission (TTC) and is made available to the public for analysis and research purposes.

4 Results

4.1 Proportion and table of Null, Zero, or Negative Values by Column

4.1.1 TTC Bus Delay Data (2014–2019)

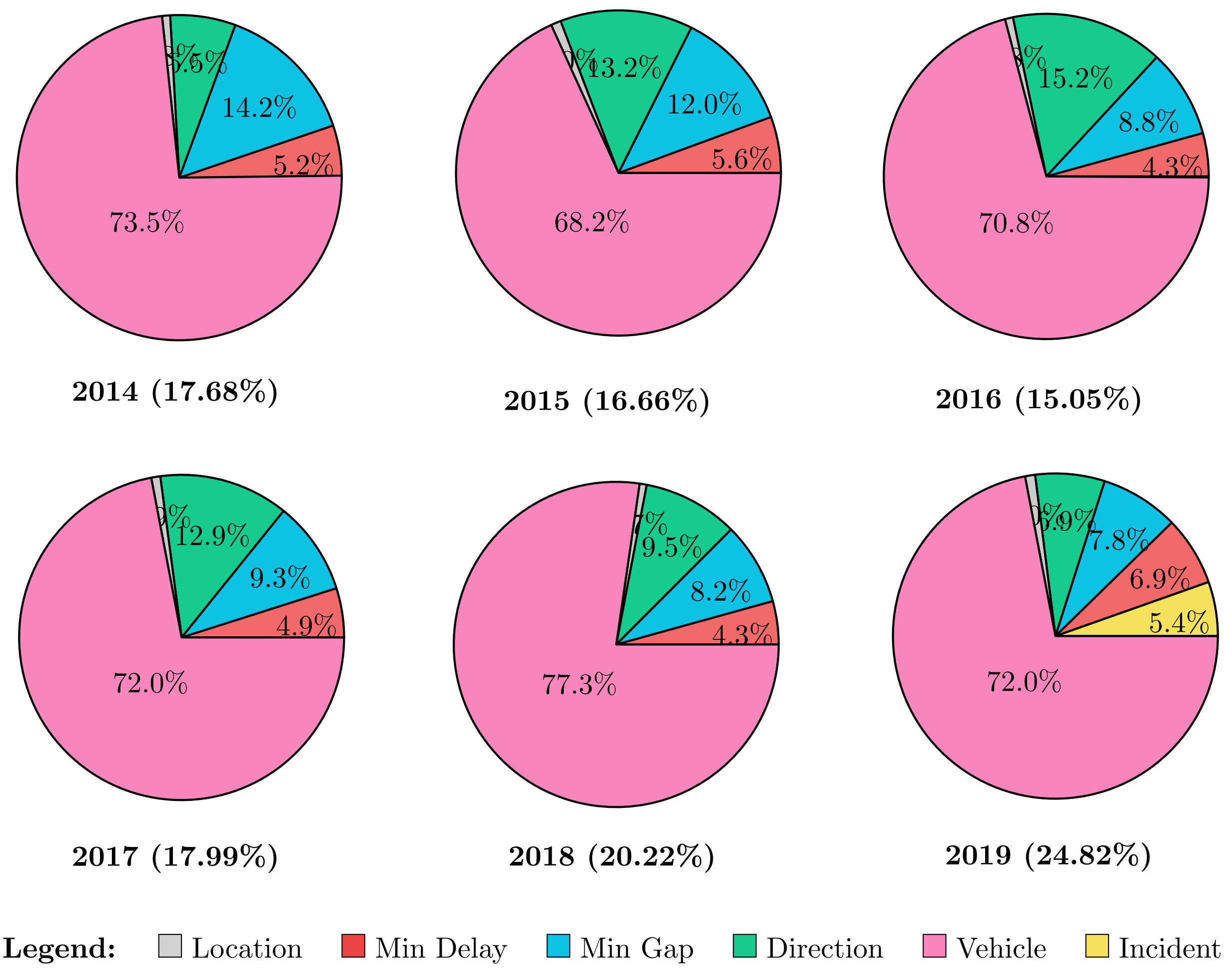
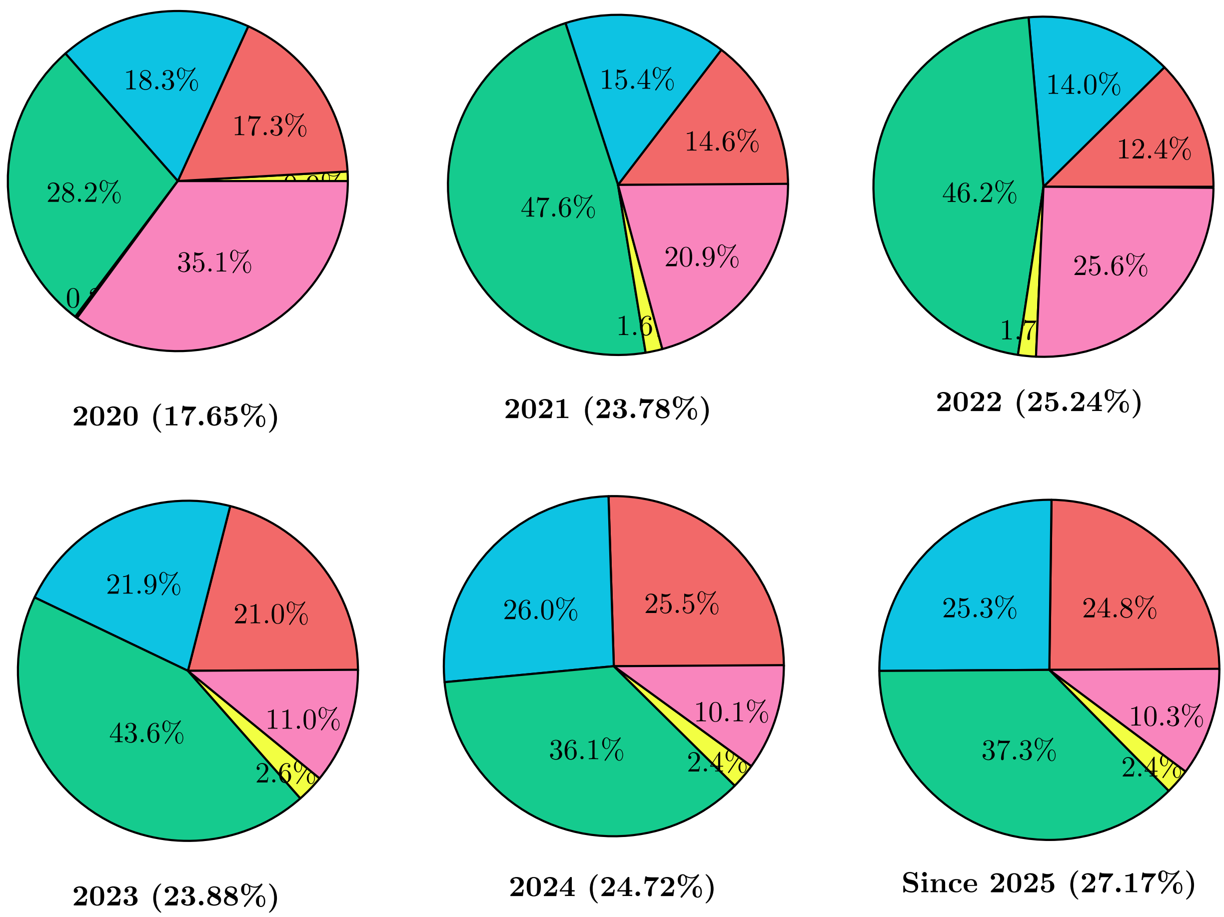


Figure 4: Proportion of null, zero, or negative values by column for buses (2014–2019).

Table 1: The number of rows with 0/negative (Z/N) or null values by column for buses (2014–2019).

	2014		2015		2016		2017		2018		2019	
	Z/N	Null	Z/N	Null	Z/N	Null	Z/N	Null	Z/N	Null	Z/N	Null
Location	1	115	0	142	0	110	0	119	0	116	4	172
Min Delay	983	20	811	3	555	2	676	7	579	123	932	254
Min Gap	2,701	34	1,732	5	1,136	8	1,299	7	890	452	974	367
Direction	2	1,242	7	1,905	1	1,960	6	1,803	1	1,554	1	1,183
Vehicle	184	13,963	31	9380	71	9,086	51	10,030	72	12,568	309	12,104
Incident	0	0	0	0	0	0	0	0	0	0	0	935
Total values	19,245		14,016		12,929		13,998		16,355		17,235	

4.1.2 TTC Bus Delay Data (2020–April 2026)



Legend: Location Min Delay Min Gap Direction Vehicle Line/Route

Figure 5: Null, zero, or negative values by column for buses (2020–April 2026).

Table 2: The number of rows with 0/negative (Z/N) or null values by column for Buses (2020–April 2026).

	2020		**2021**		**2022**		**2023**		**2024**		**Since 2025**	
	Z/N	Null	Z/N	Null	Z/N	Null	Z/N	Null	Z/N	Null	Z/N	Null
Location	0	47	0	0	0	0	0	0	0	0	0	0
Min Delay	138	4	2,242	0	2,818	0	4,590	0	6,733	0	8,710	0
Min Gap	251	7	2,369	0	3,180	0	4,801	0	6,874	0	8,881	0
Direction	2	2,518	1	7,304	0	10,480	1	9,540	1	9,531	0	13,098
Vehicle	882	2,259	3,203	0	5,802	0	2,403	0	2,657	0	3,622	0
Line/Route	0	78	0	240	0	384	0	570	5	662	0	833
Total values	8,941		15,359		22,664		21,905		26,463		35,144	

4.1.3 TTC Streetcar Delay Data (2014–2019)

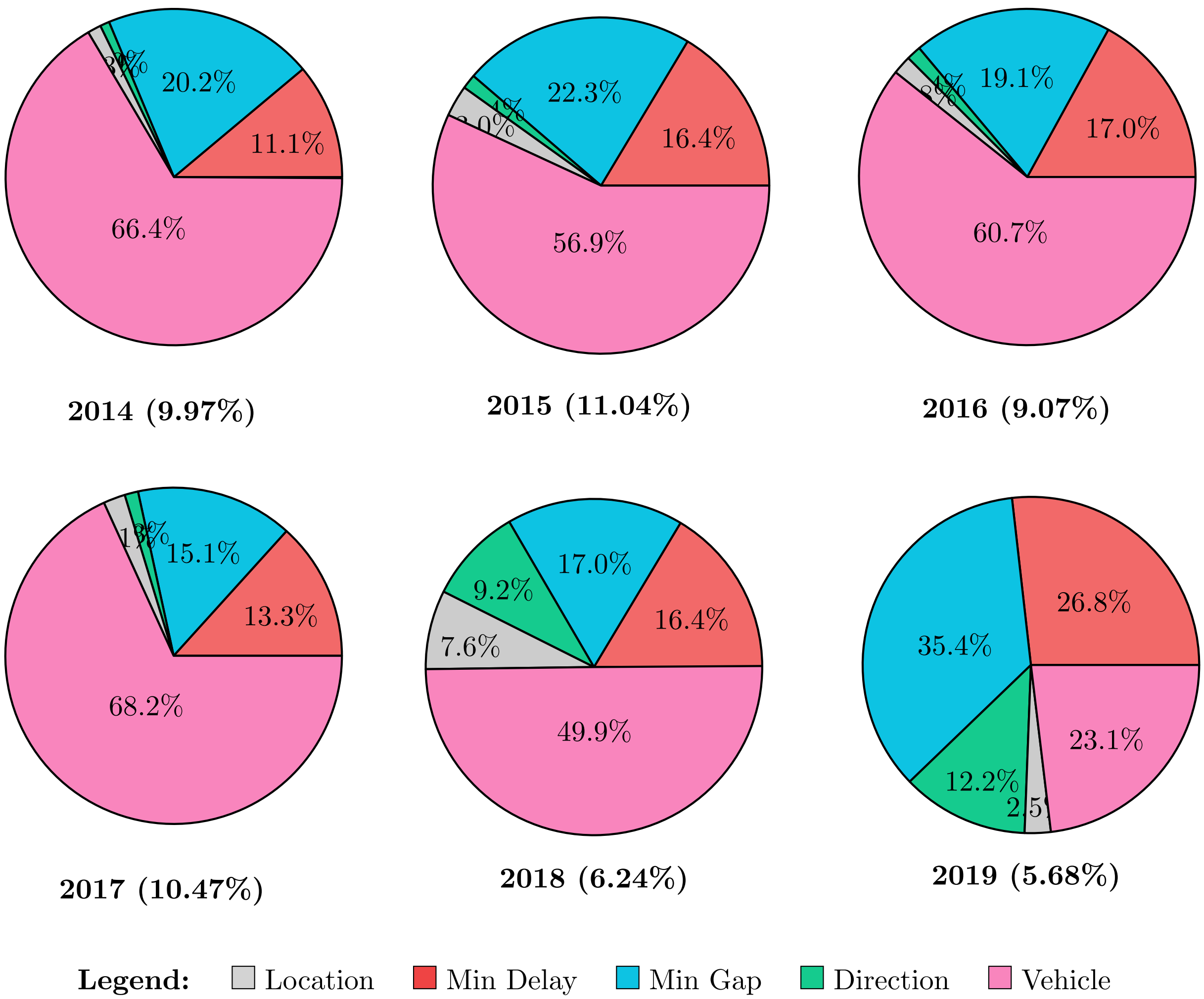


Figure 6: Null, zero, or negative values by column for streetcars (2014–2019).

Table 3: The number of rows with 0/negative (Z/N) or null values by column for Streetcars (2014–2019).

	2014		**2015**		**2016**		**2017**		**2018**		**2019**	
	Z/N	Null	Z/N	Null	Z/N	Null	Z/N	Null	Z/N	Null	Z/N	Null
Location	0	17	0	53	0	28	0	36	0	92	0	24
Min Delay	138	4	287	2	270	2	224	4	161	37	230	28
Min Gap	251	7	391	1	303	3	257	2	163	43	281	60
Direction	1	11	0	24	0	22	0	22	1	0	0	118
Vehicle	8	840	11	988	72	899	77	1,091	10	591	0	223
Total values	1,277		1,757		1,599		1,713		1,098		964	

4.1.4 TTC Streetcar Delay Data (2020–April 2026)

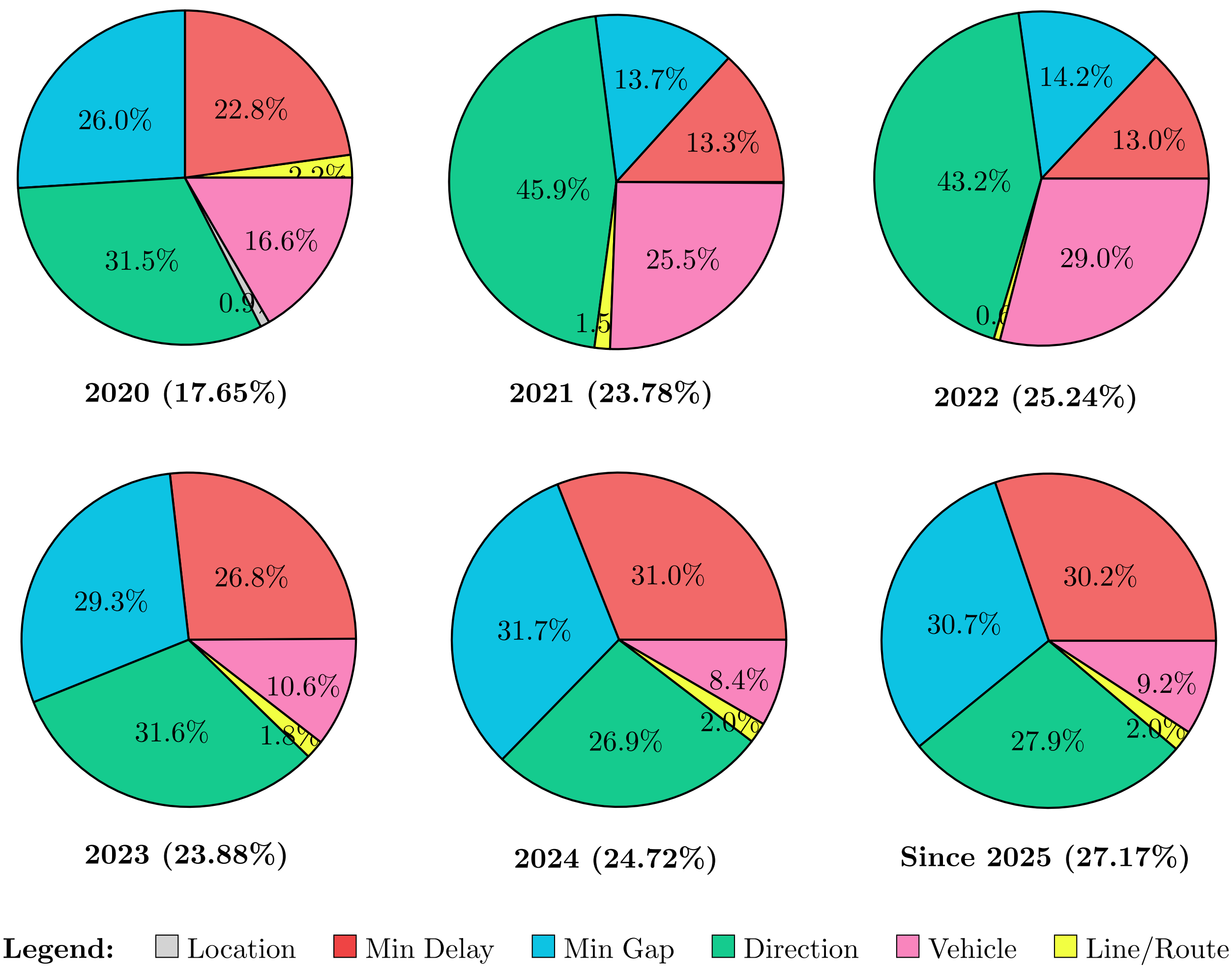


Figure 7: Null, zero, or negative values by column for streetcars (2020–April 2026).

Table 4: The number of rows with 0/negative (Z/N) or null values by column for Streetcars (2020–April 2026).

	2020		**2021**		**2022**		**2023**		**2024**		**Since 2025**	
	Z/N	Null	Z/N	Null	Z/N	Null	Z/N	Null	Z/N	Null	Z/N	Null
Location	0	14	0	0	0	0	0	0	0	0	0	0
Min Delay	355	2	790	0	915	0	1,899	0	2,455	0	3,489	0
Min Gap	401	5	810	0	1,005	0	2,075	0	2,509	0	3,557	0
Direction	0	492	0	2,726	0	3,049	0	2,240	0	2,128	0	3,231
Vehicle	210	50	1,515	0	2,050	0	749	0	666	0	1,065	0
Line/Route	0	35	0	92	0	43	0	128	0	156	0	230
Total values	1,564		5,933		7,062		7,091		7,914		11,572	

4.1.5 Interpreting and analyzing the results from section 4.1.1 to 4.1.4

Note: How to correctly interpret the pie charts?

Each pie chart shows the proportion of cells with zero, negative (Z/N), or null values in multiple columns for a given year. Each category is calculated by dividing the number of unknown values in a column by the total number of unknown values in the whole dataset.

From 2014 to 2017, both types of vehicles show a similar pattern of proportions of zero, negative (Z/N), or null values across the columns. The vehicle column shows the highest proportion of unknown values, while the line/route and location columns show the lowest proportion of unknown values. In TTC Bus Delay Data, around 70% of all the unknown values in each year's dataset are in the vehicle column. This proportion is consistent until 2019. In TTC Streetcar Delay Data, around 60% to 65% of all the unknown values are in the vehicle column. This proportion significantly decreases to 50% and 23% in 2018 and 2019 respectively. After that, the proportion of unknown values in the vehicle column in both types of vehicles is gradually decreasing, where it is around 20% to 30% in 2020 to 2023 and around 10% in 2024 to April 2026. Therefore, it can be concluded that recording the vehicle number is a manual process and may not be important for the analysis of delay data.

In the proportions of Z/N or null values that are in the Min Delay or Min Gap columns, they are roughly the same across the years, except from 2014 to 2016 in TTC Bus Delay Data and 2014 and 2019 in TTC Streetcar Delay Data. This means that it is more common for both values to be zero, negative or null in one delay event. In TTC Bus Delay Data, both columns show an increasing trend after 2019, while in TTC Streetcar Delay Data, the trend is more fluctuating across the years. This means that the data quality is decreasing since in the year 2019 and after 2022 in both types of vehicles, the proportion of unknown values in the Min Delay and Min Gap columns is significantly higher than before, around 25% to 30%.

In the proportions of Z/N or null values that are in the Direction column, both types of vehicles show a similar pattern across the years. Before 2020, the proportion of unknown values is always less than 16% in both types of vehicles. Especially in TTC Streetcar Delay Data, the proportion of unknown values is always less than 3%. After 2020, this proportion increases significantly, reaching around 25% to 30% on average and up to 48% in 2021 to 2023 in Buses and 2021 to 2022 in Streetcars. As a result, the direction of a vehicle is either poorly recorded or not important since there are several bus routes that only runs on one direction.

Finally, the proportions of Z/N or null values that are in the Line/Route column or Location column only appear in the smallest proportion of unknown values in all the columns. This means that there has been some error in recording the events, since these values are easy to record and check. Additionally, these 2 values can be easily recorded automatically by the system.

4.2 Proportion of rows with at least one zero, negative, or null value by year

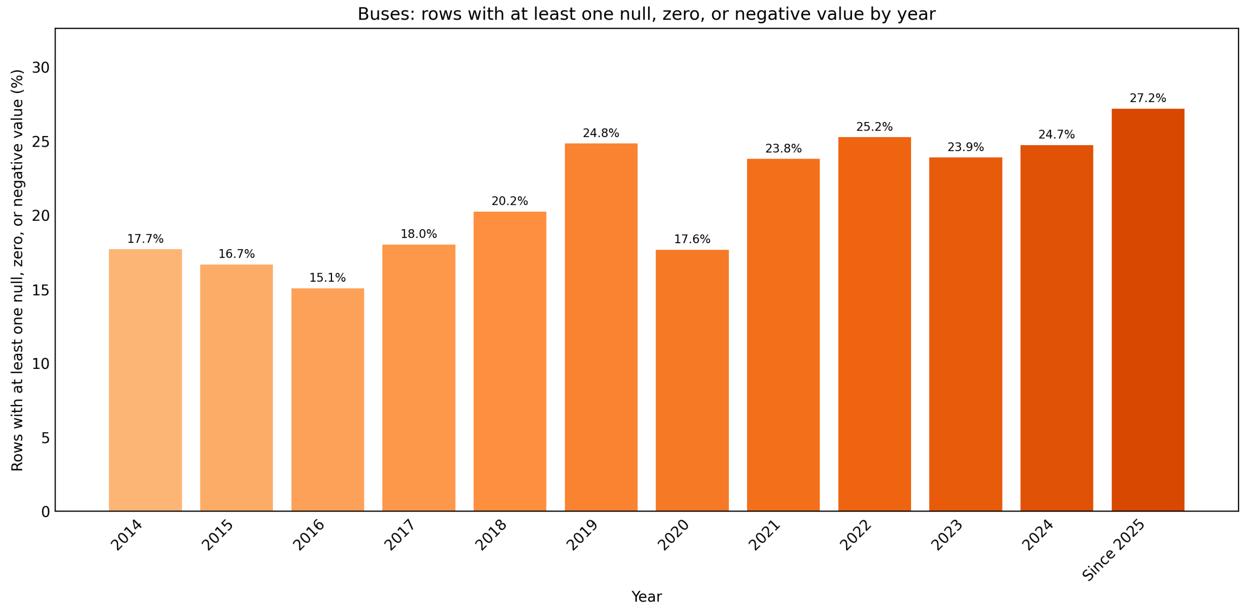


Figure 8: Yearly percentage of rows with at least one zero, negative, or null value in TTC Buses.

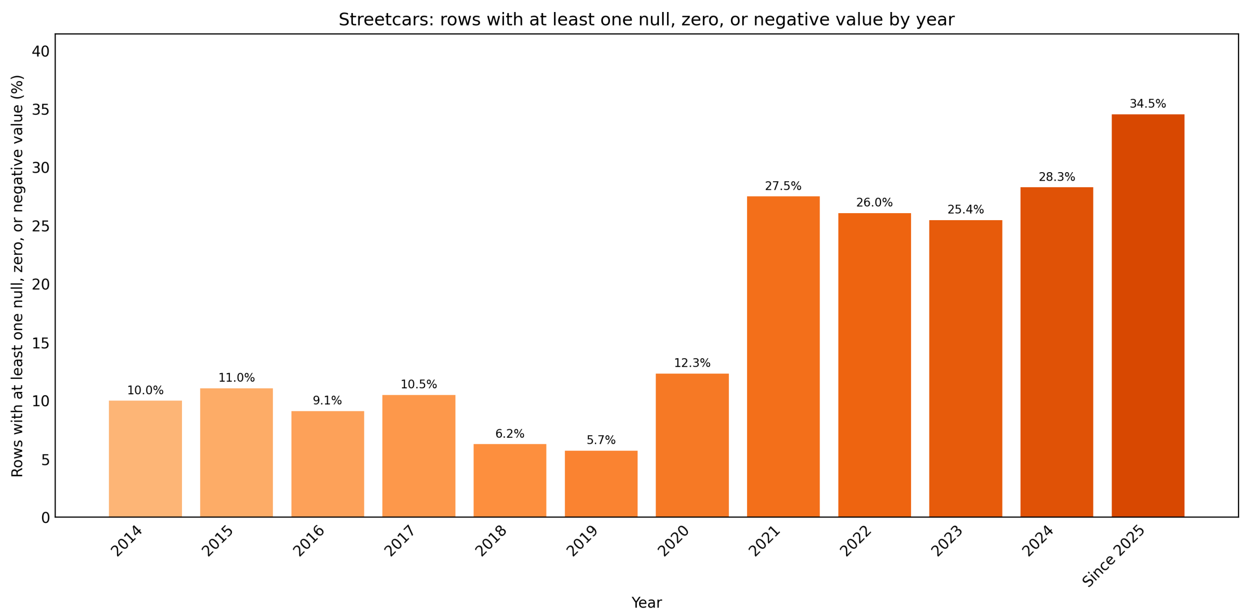


Figure 9: Yearly percentage of rows with at least one zero, negative, or null value in TTC Streetcars.

Both bar graphs show a significant increase in the percentage of rows with at least one zero, negative, or null value in the later years. In TTC Bus Delay Data, the percentage is slowly increasing throughout the years, and the latest year has the highest percentage of 27.2%. In TTC Streetcar Delay Data, the percentage of rows with at least one unknown value is relatively low from 2014 to 2020, mostly between 5% and 12%, then rises after that, which is between 25% and 30%.

4.3 Proportion of rows with a generic ratio of Min Delay to Min Gap

Our observations

Besides noticing a common ratio of Min Delay to Min Gap as 1 : 2, we also noticed that the ratio is somtimes 1 : 1 or 1 : 3. The line graphs below show the percentage of rows with a generic ratio of Min Delay to Min Gap out of total rows recorded in each year.

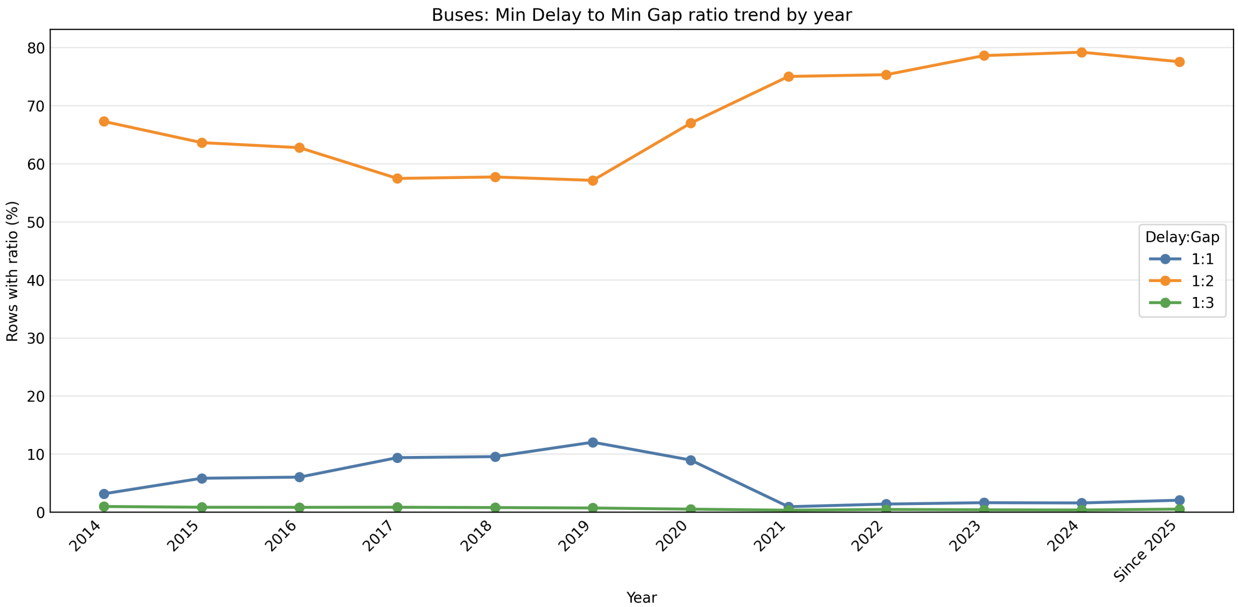


Figure 10: Yearly percentage of rows with a generic ratio of Min Delay to Min Gap in TTC Buses.

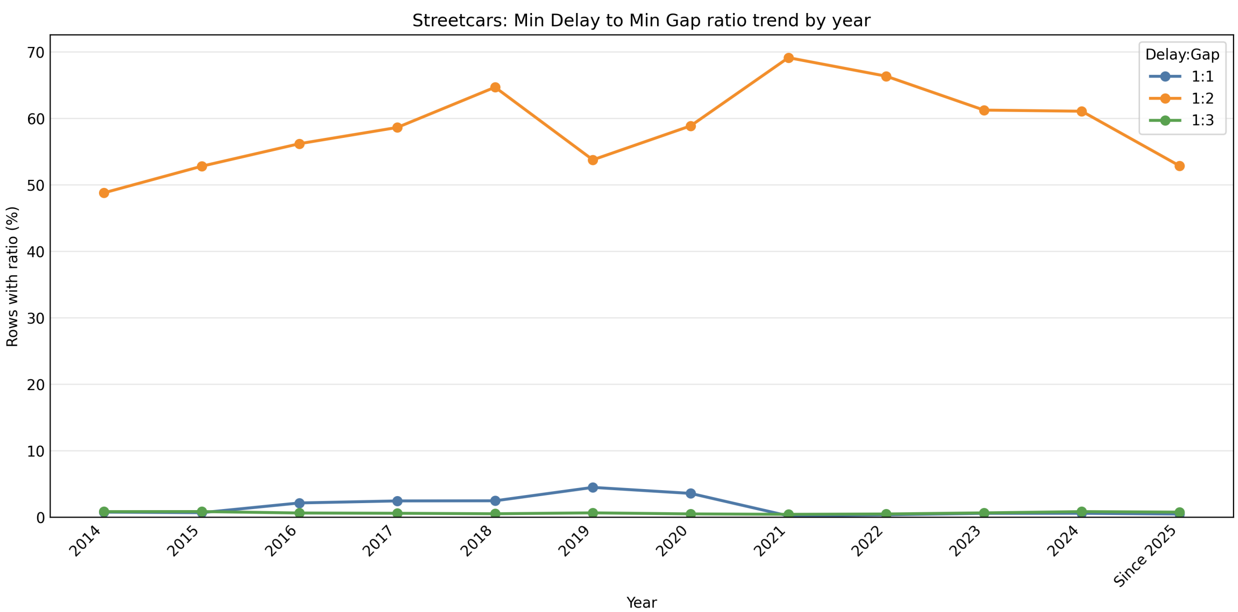


Figure 11: Yearly percentage of rows with a generic ratio of Min Delay to Min Gap in TTC Streetcars.

Both line graphs show the majority of ratio 1 : 2 in both TTC Buses and Streetcars delay data. In TTC Buses, the percentage of rows with this ratio 1 : 2 varies from 60% to 80% throughout the years, while it varies from 50% to 70% in TTC Streetcars. On the other hand, the percentage of rows with ratio 1 : 1 or 1 : 3 is significantly low, mostly less than 10% in both types of vehicles. On average, the percentage of rows with ratio 1 : 1 is around 3% in TTC Streetcars and around 4% in TTC Buses and ratio 1 : 3 is around 1% in TTC Streetcars and around 2% in TTC Buses.

5 Limitations

Through our analysis, we have identified several limitations in the TTC delay data. First of all, we only collected public dataset from Toronto Open Data Portal, which is a public dataset portal for the City of Toronto. Therefore, the data might not be comprehensive for the public and there might not be a proper process to collect all the data from the TTC. Second, the data is not consistent across the years. For example, in TTC Bus Delay Data on July and August 2021, all the delayed events recorded are for the Streetcars, and they are identical to the Streetcars delay data on July and August 2021. This means that the data is not always accurate and there might be some errors in the data. Finally, there are 2 verions of column names for recording the minutes of delay and gap, which are Min Delay & Min Gap and Delay & Gap. The second version appeared in 2020, which raises a question about whether the minutes recorded in Delay & Gap have the same meaning as the minutes recorded in Min Delay & Min Gap.

6 Conclusion

6.1 Overall Data Quality Assessment

The data quality is poor and worsening. Across the full 2014–April 2026 period covering 776,446 bus and 163,010 streetcar delay events, the proportion of records containing at least one zero, negative, or null value has grown from approximately 15–18% in the mid-2010s to 27.2% for buses and 34.5% for streetcars in the most recent data. No corrective trend is visible in either series after 2019. The root cause is structural: the current system relies on operators entering free-form text with little or no automated validation, and the data schema has been revised multiple times without backward-compatible versioning or a published changelog. The dataset is not ready for predictive modelling and requires substantial cleaning before even descriptive analysis is reliable.

6.2 Proposed Data Collection and Management Process

The core principle is that any field a machine can record automatically should never require human text entry. Therefore, we propose the following data collection and management process:

6.2.1 Track 1: Automatic Data Capture from On-Board Sensors

TTC buses and streetcars already carry Computer-Aided Dispatch and Automatic Vehicle Location (CAD/AVL) systems, GPS receivers, and door-open sensors. These systems should automatically populate the following fields without any operator input.

- **Date and Time:** Sourced from the system clock at the moment the delay event is created.
- **Route and Vehicle ID:** Pulled from the CAD/AVL assignment at the time of the event.
- **Location:** Resolved from the GPS coordinates to the nearest GTFS stop ID, providing a stable, unambiguous reference rather than a free-text location name.
- **Direction:** Derived from the trip pattern registered in the CAD/AVL system, constrained to the four valid values: N/B, S/B, E/B, W/B.
- **Min Delay:** Calculated automatically as the difference between the scheduled departure time and the actual departure time, both of which CAD/AVL records.
- **Min Gap:** Derived from the real headway to the following vehicle, also available from AVL. This field must never be a manual entry or a formula of the delay value; it should reflect an independent measurement.

6.2.2 Track 2: Constrained Operator Input

Only fields that genuinely require operator judgment should be entered manually. The operator interface should be a mobile application presenting the following two inputs only. In addition, an optional free-text notes field may be included for supplementary context, but its contents must not be stored in any column used for quantitative analysis.

- **Incident type:** A dropdown menu populated with the current approved vocabulary. Free-text entry must not be permitted. The vocabulary should be versioned (see Section 6.2.5).
- **Start/End Delay:** The operator taps "start delay" when an event begins and "end delay" when it clears. This provides a manual timing cross-check against the AVL-derived figure and requires no text entry.

6.2.3 Validation Layer

Before any record is written to the database, an automated validation step must enforce the following rules. Records failing any of these checks must be routed to a reject queue for supervisor review. They must not be silently written to the database with invalid values. Therefore, the validation layer is a crucial step to ensure the data accuracy.

- `Min Delay` ≥ 0 and `Min Gap` ≥ 0.
- `Direction` is one of N/B, S/B, E/B, or W/B.
- `Route` and `Vehicle` are non-null integers matching the current fleet and route registry.
- `Incident` is a value from the currently approved vocabulary.
- The record is not a duplicate of an existing event within five minutes on the same route.

6.2.4 Central Database and Unique Event Identifier

Every confirmed event must be assigned a system-generated unique identifier (`EventID`) at write time. This is the single most important structural change: it makes deduplication unambiguous, enables external systems to reference specific events, and allows corrections to be linked to their originals without overwriting history. The database should be an append-only log; corrections are new records linked to the original by `EventID`, not in-place overwrites.

6.2.5 Schema Versioning and Open Data Release

Column names, data types, and the incident vocabulary must be treated as a versioned schema. Any change to these elements increments the schema version number and is published in a human-readable changelog alongside the corresponding data release. Consumers of the data can then detect schema changes programmatically and update their pipelines accordingly.

As an immediate remediation, the column names should be standardised to `Min Delay` and `Min Gap` as the canonical form across all years, with the 2020-onward data backfilled to use these names.

6.2.6 Nightly Data Quality Audit

An automated job should run each night over the previous day's records and flag the following conditions for supervisor review.

- Records where `Min Gap` = 2 × `Min Delay` (suggesting formula fill rather than independent measurement).
- Records where `Min Delay` = 0.
- Routes with a null rate in `Direction` exceeding a configurable threshold (e.g., 10%).
- Any `Incident` value not present in the current approved vocabulary (catching typographical variants and ad-hoc additions before they propagate).

Flagged records are surfaced in a supervisor dashboard rather than removed. The raw history is preserved; the flags are stored separately and linked by `EventID`.

7 Link to our source code repository for the working prototype

The GitHub repository of our source code for our working prototype is available at
github.com/dqhung711/the_delayticans_tdc_2026.